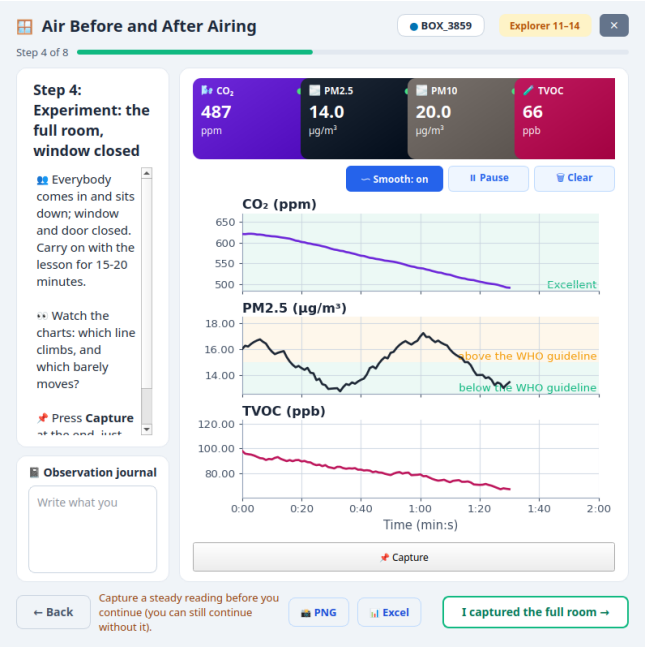
Air Before and After Airing

Three sensors on one box — and the question of whether they tell the same story

Level: Explorer and Scientist — ages 11–19 (Romanian school years V–XII) • **Time in class:** one 50-minute lesson; a two-lesson project with a logging day • **Sensors:** SCD41 (CO_2), BMV080 (particles), SGP30 or SGP41 (vapours) — any one of them



The activity screen, with a box carrying all three sensors. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer and Scientist — ages 11–19 (Romanian school years V–XII). In years V–VI pupils compare curve shapes and read thresholds; from year VII they add where each quantity comes from (biology: respiration; chemistry: volatile organic compounds), and at upper secondary level the discussion of fine particles and health (geography: urban air pollution).

Assumed knowledge

Reading a graph over time and subtracting two values. The idea that air is a mixture. Nothing about particles or organic compounds: the activity introduces them.

Time in class

One 50-minute lesson: the baseline during the break, the full room for 15–20 min, the window open for 10 min, the analysis. Project: a second lesson for a day of logging and for comparing days with different weather.

Teacher preparation

15 min — switch the box on at least ten minutes beforehand: the vapour sensors need a minute to warm up, and the SGP41's VOC index learns the room's air during its first hours. **Check which sensors the box has**: the app lists them on the **Home** page. Decide where the box will stand — at the head height of a seated pupil, at least a metre from anyone, **with nothing above the particle sensor's lens for about 35 cm**.

Sensor settling time	CO_2: 30 seconds after power-up, then one value every five seconds. Particles: nothing for the first 15 seconds, then one value a second. Vapours: 15 seconds (SGP30) or about a minute (SGP41). **For the baseline, two to three minutes** of settling in the aired room.
Sensors and boxes	Any of the air sensors of firmware 2.4 (the EASE summer school): the **SCD41** for CO_2, the **BMV080** for PM2.5 and PM10 particles, the **SGP30** (TVOC, in ppb) or the **SGP41** (the VOC index). The lesson works with one, two or all three; with all three the discussion is at its best. **Classroom boxes on firmware 2.3.0 or 2.2.6, with only the usual sensors, cannot run the lesson**, except those with an SCD41: they run it for CO_2 alone, and for them *Indoor Pollution* is the better lesson. The "particle sensor blocked" warning needs firmware 2.4.4.

Three routes

Short	One 50-minute lesson: the prediction (5), the baseline in the aired room (5), the full room with the window closed (15–20), the window open (10), the analysis and conclusion (10). This is the wizard as written.
Full	Short version, 30 min: the baseline is taken outdoors or in an empty corridor, the full room lasts 10 min, the open window 5 min. It works, but differences in particles are harder to see.
Project	Two-lesson project: a whole day of logging with a paper log (who is in the room, when the window is opened, what is outside), then a lesson in which pupils match the peaks to the log and compare a clear day with one of fog or of wood-burning stoves in the neighbourhood.

The application starts in English. Buttons are named here exactly as they appear on the screen.

The lesson is hidden: switch it on once in Settings

The lesson needs sensors that the usual classroom boxes do not have, so the app does not show it by default. Open **Settings**, go to the **Visible activities** section and tick *Air Before and After Airing*; the lesson appears at once on the **Activities** page. The tick stays on that computer.

1. Where this fits in the curriculum

- **Geography** — in year V, the composition of air; in year IX, air pollution and its sources. Fine particles are the urban pollutant with the largest effect on health, and here pupils measure them in their own classroom.
- **Biology** — in year VII, human respiration: the CO_2 in the room comes from the pupils' lungs; at upper secondary level, the effect of fine particles on the airways.
- **Mathematics** — in year VI, ratios and comparing two values; at upper secondary level, reading a noisy series and its mean.
- **Physics** — in year VIII, light and its scattering: the particle sensor counts specks of dust by the light they scatter out of a laser beam.

The core skill is **comparing three different measurements of the same thing**. Each sensor responds to a different part of the air, and pupils learn that "clean air" is not a single number.

2. Time plan

Route	Time	What happens
One lesson	**50 min**	The prediction (5), the baseline in the aired room (5), the full room with the window closed (15–20), the window open (10), the analysis and conclusion (10).
Short	30 min	The baseline outdoors or in the corridor, the full room for 10 min, the window for 5 min.
Between lessons	one school day	The box logs in the classroom, with a paper log on the wall: how many people are in the room, when the window is opened, what is outside (traffic, fog, smoke from stoves).
Lesson 2	50 min	Open the day's export on the **Data & Export** page and match every rise and every fall to the log.

3. Materials

Item	Per group	Notes
MagicBox with at least one of the SCD41, BMV080, SGP30, SGP41	1	The box has three sockets; a box with CO_2, particles and vapours fills all three.
Laptop with the app	1	Over USB or WiFi; the lesson has to be ticked once in **Settings** → **Visible activities**.
A room with a window that opens	1	The lesson depends on opening and closing it at the moment you choose.
Stopwatch or phone	1	For the 15–20 minutes of the full room and the 10 with the window open.
Masking tape	a few strips	To mark where the box stands.

Preparation the day before. Switch the box on and watch each value for a few minutes on **Live Data**: outdoors or at an open window the CO_2 should sit around 400–450 ppm, and in a quiet room the particles usually stay below 20 $\mu g/m^3$. Hold your hand about ten centimetres above the particle sensor: the app should show the warning that the sensor is blocked. That tells you the firmware is 2.4.4 and that pupils will see the message if they cover the sensor.

4. Scientific background for the teacher

Three sensors, three parts of the air

Carbon dioxide (SCD41). We breathe it out: a seated pupil produces about 0.2 L a minute. Thirty pupils in a 170 m^3 room raise the CO_2 by some 30 ppm a minute. At classroom levels CO_2 is not toxic; it is a **ventilation marker**, showing how much of the air has already been through somebody's lungs. Its source lies almost entirely *inside* the room.
Particles (BMV080). PM2.5 and PM10 are the mass of particles smaller than 2.5 and 10 micrometres in a cubic metre of air, in $\mu g/m^3$. The sensor counts the specks that cross a laser beam above its lens and estimates their size; Bosch's software inside the sensor then turns the count into a mass. Particles come **from both sides of the window**: indoors,

footsteps, clothes, chalk and dust stir them up; outdoors, they come from traffic, stoves and boilers, building sites and pollen. That is why, when you open the window, the particles may fall *or* rise: it depends on the air outside.
Vapours (SGP30 or SGP41). Volatile organic compounds are the gases given off by markers, glue, varnish, perfume, deodorant and cleaning products, and by our own bodies. The SGP30 estimates their total, **TVOC**, in ppb (parts per billion). The SGP41 gives a **VOC index** from 1 to 500, where 100 is the room's average over the last few hours: above 100 the air holds more vapours than usual, below 100 fewer. As with CO_2, their source is mostly inside the room.

4.1 The thresholds and what they mean

Quantity	Reference	What it means
CO_2	the app's bands: below 800 ppm *Excellent*, 1200–2000 *Moderate*	Rounded from the European ventilation standard EN 16798-1. They show ventilation, not pollution.
PM2.5	WHO: 15 $\mu g/m^3$ as a 24-hour mean	The World Health Organization's 2021 guideline. The recommended annual mean is 5 $\mu g/m^3$.
PM10	WHO: 45 $\mu g/m^3$ as a 24-hour mean	The same guideline; the recommended annual mean is 15 $\mu g/m^3$.
TVOC (SGP30)	no official threshold	The value depends on the mix of gases; compare the phases with one another, not with a number from elsewhere.
VOC index (SGP41)	100 = the room's usual air	A comparison with the recent past of the same room, not a concentration.

Be careful with pupils on one point: **the WHO thresholds are averages over a whole day**. A reading of 25 $\mu g/m^3$ for ten minutes does not "exceed the guideline"; it only shows that the air was more loaded during those minutes. Within a single lesson the fair comparison is between the phases: before, with the room full, after airing.

5. Lesson flow

The in-app wizard has eight steps: the introduction, the prediction, the baseline in the aired room, the full room with the window closed, the open window, the analysis, the **How sure are we?** step and the conclusion. The screen shows a card and a chart only for the quantities the box has; if a sensor is connected later, its card appears by itself.

1. **The written prediction (5 min).** *When we open the window, which quantities will fall? Could one stay the same, or even rise?* Most classes say that everything falls. The app keeps the prediction and quotes it back in the conclusion.
2. **The introduction on screen (2 min).** The three sensors, in plain words. The activity opens from the **Activities** page with **Start**; the step ends with **Got it!**

Technical note — what the screen shows and what is saved

The cards and the charts. CO_2 in ppm, PM2.5 and PM10 in $\mu g/m^3$, TVOC in ppb or the VOC index without a unit. PM10 has a card but no chart of its own: it follows PM2.5, since it comes from the same sensor, and four charts do not fit on a laptop screen. The CO_2 chart carries the app's bands; the PM2.5 chart has a green band below the WHO guideline

and an orange one above it.
The Capture button. It saves the mean of the last five seconds for *all* the box's quantities at once. Press it once per phase: at the baseline, at the end of the full-room phase (just before opening the window) and after ten minutes with the window open. The green button changes its label: **I captured the aired room**, **I captured the full room**, **I captured the aired room again**, **I analysed the data**, **Continue** and, at the end, **Finish**.
The particle sensor blocked. If something still sits less than about 35 cm above the lens (a lid, a book, a shelf, a hand), the sensor stops measuring and the app shows a warning window that blocks nothing and closes by itself once the sensor sees clearly again. A reading captured in that time has no particles; the table shows a dash.
What is missing at first. Particles are absent for the first 15 seconds after the box is switched on, vapours for 15 seconds (SGP30) or about a minute (SGP41). The VOC index needs a few hours to learn the room: in the first hour after power-up it can stay below 100 even in ordinary air. Compare the phases with one another, not with yesterday.
Exporting. PNG and **Excel** appear at the measurement steps. Every quantity is also written to the box's CSV log, with its unit.

3. **The baseline, in the aired room (5 min).** The room aired during the break, the window closed, the box in its place. Wait for the timer on the screen and press **Capture**. Note how many people are in the room.
4. **The full room, window closed (15–20 min).** Everyone comes in and the lesson goes on. The CO_2 climbs visibly; the vapours climb mostly if someone uses markers or perfume; the particles jump when pupils move about and settle again when they sit still. At the end, **Capture**.
5. **The window open (10 min).** Open the window and, if you can, the door on the opposite side. Note what is outside: a busy street, a yard, a park, fog. After ten minutes, **Capture**.
6. **Analysis and conclusion (10 min).** The app sets out the three phases for each quantity in a table, with the change from the full room to the open window and with the baseline. The class answers: what fell, what did not, and why.

The investigation, in one table

The question. Do all the sensors say the same thing about the air in our classroom, before and after airing?
The hypothesis. Written before measuring, for each sensor on the box: does it rise with the room full? does it fall with the window open?
What I change. The number of people in the room, then the state of the window.
What I measure. Every quantity the box has, with one reading captured at the end of each phase, and what is outside.
What I keep the same. Where the box stands, nobody within a metre of it, nothing above the particle sensor.
How many repeats. The same lesson on a day with different weather — clear, foggy, windy — is the best repeat: outdoor particles change a great deal from one day to the next, CO_2 hardly at all.
What makes a reading valid — and what makes it an artefact. A pupil breathing over the box raises the CO_2 and the vapours; a book left over the particle sensor stops the measurement; someone shaking a board duster next to the box produces a particle peak that does not belong to the room.

6. Analysis after the activity

6.1 What the data should look like

Quantity	What you usually see	Why
CO_2	rises a lot with the room full, falls quickly on airing	The source is in the room; outdoors there are about 420 ppm.
TVOC or the VOC index	rises a little or a lot, depending on what is used in the room; falls on airing	The source is mostly in the room.
PM2.5 and PM10	jump when pupils move; on airing they fall, stay or rise	They come from both sides; it depends on the air outside.

The orders of magnitude are only a guide: CO_2 between 500 and 2000 ppm, PM2.5 from a few units to a few tens of $\mu g/m^3$, the VOC index between 50 and 300. **Do not look for a "correct" number**; look at whether each quantity went up or down, and by how much compared with the others.

6.2 Questions to ask afterwards

1. Which quantity changed most when the class came in? Which changed least?
2. Did the particles fall or rise with the window open? What was outside at that moment?
3. Why can we not bring the CO_2 below the outdoor level, however much we air the room?
4. The ten-minute PM2.5 value was above 15. Can we say that the room exceeds the WHO guideline? Why not?
5. What would you do on a day of fog and heavy traffic: air the room, or not?

6.3 Assessment ideas

- **Low stakes:** the pupil completes the table of the three phases and says, for each quantity, whether it went up or down.
- **Standard:** the pupil explains where each quantity comes from and why airing affects them differently.
- **Extension:** a ventilation recommendation that also takes the outdoor air into account, with a stated limitation of the measurement.

Common pitfalls and misconceptions

"Airing cleans everything." It cleans out what is produced in the room. If the air outside carries more particles, the open window brings them in — and that is the most valuable conclusion of the lesson, not an error.
Comparing ten minutes with a 24-hour threshold. The WHO thresholds are daily means. Within one lesson, compare the phases with each other.
An object above the particle sensor. The sensor stops measuring, and the app says so. Move the object, not the box.
The VOC index read as a concentration. An index of 150 does not mean "one and a half times more vapour than is allowed" but "more than usual in this room".
The SGP30's eCO_2 read as CO_2. The SGP30 also gives an estimate of CO_2, calculated

from the vapours. It is not a CO_2 measurement; if the box has an SCD41, the SCD41 is the one that counts.
Movement next to the box. Footsteps, clothes being shaken and the board duster raise dust right beside the sensor. Keep the box away from the board and from the coat rack.

Safety and classroom management

Windows are opened by the teacher; nobody climbs on a sill, a radiator or a chair.
Cold. Air briefly and fully; move pupils away from the window while it is open.
The particle sensor's laser is enclosed in the sensor's housing and cannot be seen; even so, do not look closely into the lens and do not take the sensor apart.
Sprays and perfumes. If you want to show the vapours rising, use an open marker next to the box, not a spray: some pupils have asthma.

Adaptations for pupils with special educational needs

One quantity per pupil. Pupils who tire quickly of tables follow a single quantity — for example only the CO_2 — and say at each phase "up", "down" or "the same". The class's conclusion is built from all of them.
Roles. One pupil times the phases, one notes what is outside, one reads the cards aloud.
Asthma and allergies. No sprays; an open marker is enough to show the vapours.
Cold and draughts. Sensitive pupils move away from the window before the airing.

If things do not go to plan

The box has only one sensor. The lesson works with it. With CO_2 alone, *Indoor Pollution* is the better choice, since it also computes the rates.
You cannot air the room during the break. Take the baseline outdoors or in the corridor and tell the class that you did.
The particles do not change at all. This happens on days with clean outdoor air and a quiet class. The result is valid: say so. For a visible difference, let a pupil shake a board duster a metre from the box, as a separate demonstration.
The blocked-sensor warning does not go away. Look above the box up to about 35 cm: a shelf, a cupboard, a lamp. Move the box to a spot open to the ceiling.
The box will not connect. Use the "What the data should look like" table as a printed dataset and discuss it with the class.

7. Further resources

- **World Health Organization** — `https://www.who.int` — search for *WHO global air quality guidelines* (2021), with the values for PM2.5 and PM10.
- **European Environment Agency** — `https://www.eea.europa.eu` — the map of air quality across Europe, including the stations in Romania.
- **Romania's national air quality monitoring network** — `https://www.calitateaer.ro` — the outdoor values in your town, to set beside those in the classroom.
- **REHVA** — `https://www.rehva.eu` — classroom ventilation and the standard EN 16798-1.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *if there is fog and traffic outside, do you open the window or not? Why?*

For a more formal judgement, use the **common three-axis rubric**:

- **Measurement practice** — the pupil takes the baseline in the aired room, captures a reading at the end of each phase, notes what is outside and keeps the particle sensor clear.
- **Data interpretation** — the pupil compares the phases for each quantity and does not confuse a ten-minute value with the WHO daily mean.
- **Scientific reasoning** — the pupil explains why airing affects CO_2, vapours and particles differently, and states a limitation of the measurement.

The rubric itself, with descriptors for all levels, is defined in the *curriculum plan*.

9. Pupil sheet

The sheet is a separate document, in the folder `docs/en/student_sheets`: the file `air_before_and_after_airing.pdf`. It fits on one sheet, printed on both sides, built on the three layers: what I see on the screen, what I do on my own, the conclusion I write. The table of the phases has rows for all five quantities; pupils fill in only the rows for the sensors their box has and cross out the rest.